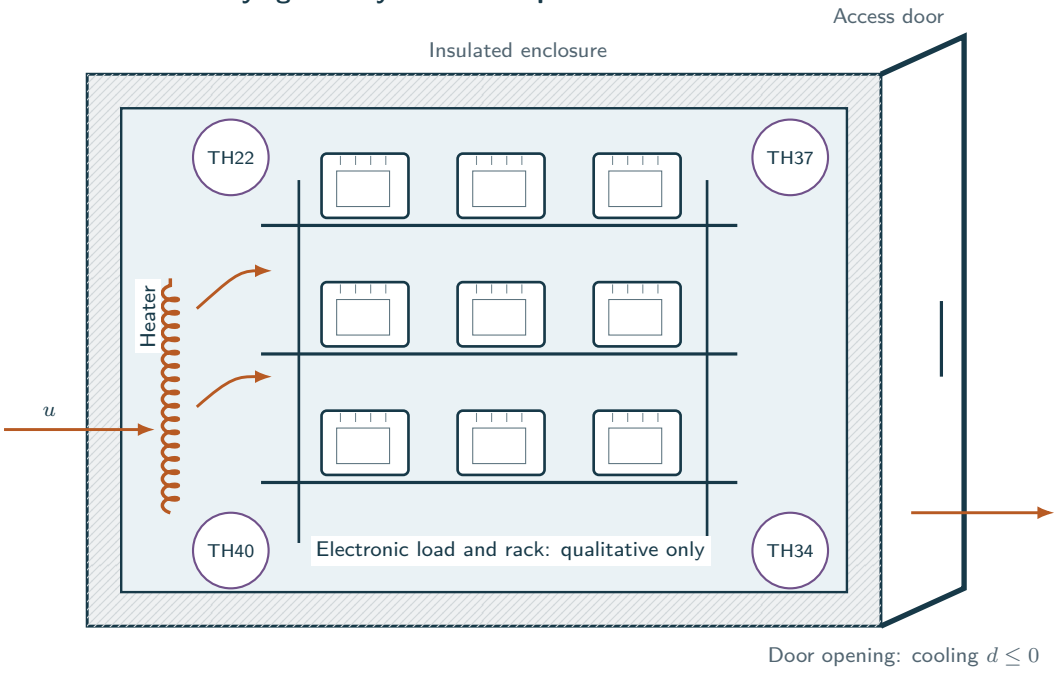
Burn-in chamber: physical picture and reduced model

01 / Chamber, four measurement channels and the meaning of the FOPDT approximation

Illustrative cutaway: geometry and sensor positions are not recovered from the thesis

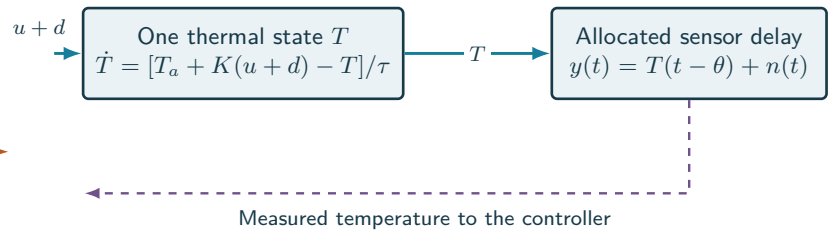


What the identified source model represents

Combined heating / measurement response:

$$G(s)H(s) = \frac{\bar{K}\, e^{-\theta s}}{\bar{\tau} s + 1}.$$

Mean-temperature parameters: $\bar{K} = 7.624$, $\bar{\tau} = 151.6$ s, $\theta = 8.78$ s. The effective command is not calibrated in watts.



A separate four-channel synthetic dataset

TH22, TH37, TH40 and TH34 are named channels; their mean and spread can be inspected. The one-state feedback simulation does not resolve a spatial temperature field. Mean-temperature tracking alone cannot certify uniformity.

Physical interpretation boundary

The cabinet, rack, heating element and sensor locations explain thermal and measurement paths; they are not recovered hardware dimensions, wiring or a safety design. No actual equipment layout or measured temperatures are supplied by this illustration.

Declared simulation choices

Ambient $T_a = 30\,^{\circ}\mathrm{C}$; reference $65\,^{\circ}\mathrm{C}$; effective command $0 \leq u \leq 10$; baseline sample period 1 s. Delay allocation to the sensor, door-loss amplitude and four-channel field generation are assumptions, separate from source-reported FOPDT parameters.

Blue arrows: model/signal propagation. Orange arrows: qualitative heating/cooling paths. Purple sensors: named measurement channels; their placement is illustrative. See source_traceability.md and models.py.

Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.